

05_Product_EDA

Q1. How Many Products Are Available in the Marketplace?

Result: 32,951

Insight: The marketplace offers 32,951 products, indicating a diverse product catalog capable of serving a wide range of customer needs across multiple categories.

Q2. How Many Product Categories Exist in the Marketplace?

Result: 73 categories

Insight: The marketplace product catalog spans 73 categories, demonstrating substantial product diversity and enabling the platform to serve a wide variety of customer segments and purchasing needs.

Q3. Which Product Categories Have the Highest Number of Products?

Result:

cama_mesa_banho : 3,029
esporte_lazer : 2,867
moveis_decoracao : 2,657
beleza_saude : 2,444
utilidades_domesticas : 2,335
automotivo : 1,900
informatica_acessorios : 1,639
brinquedos : 1,411
relogios_presentes : 1,329
telefonica : 1,134

Insight: Home and lifestyle categories dominate the marketplace catalog. Categories such as bed, bath & table, sports & leisure, furniture & decoration, and beauty & health contain the largest product assortments, indicating strong seller participation and broad customer demand in these segments.

Q4. Which Product Categories Have the Smallest Number of Products?

Result:

CDs, DVDs & Music : 1
Insurance & Services : 2
PC Gaming : 3
Kids & Youth Fashion : 5
Home Comfort 2 : 5
Tablets, Printing & Imaging : 9
Mattresses & Upholstery : 10
La Cuisine : 10
Kitchen Appliances & Food Preparers : 10
Baby Care & Hygiene : 12

Insight: Several categories contain very few products, indicating niche offerings or limited seller participation. These categories contribute little to overall catalog diversity and may represent specialized market segments with lower demand.

Q5. What Is the Average Product Weight?

Result: 2,276.47 grams (2.28 kg)

Insight: The average product weight is approximately 2.28 kg, indicating that the marketplace primarily deals with lightweight to medium-weight products. Product weight is an important factor influencing shipping costs, logistics planning, and delivery efficiency.

Q6. What Are the Average Product Dimensions?

Result:

Average Length : 30.82 cm
Average Height : 16.94 cm
Average Width : 23.20 cm

Insight: The average product sold on the marketplace measures approximately 31 cm in length, 17 cm in height, and 23 cm in width. These dimensions indicate that most products are relatively compact, which helps optimize storage, packaging, and transportation costs.

Q7. Which Categories Have the Highest Average Product Weight?

Result:

Mattresses & Upholstery : 13,190 g
Office Furniture : 12,740.87 g
Kitchen, Dining & Garden Furniture : 11,598.56 g
Bedroom Furniture : 9,997.22 g
Home Appliances 2 : 9,913.33 g
Living Room Furniture : 8,934.85 g
PCs : 7,995.33 g
Industry, Commerce & Business : 5,929.19 g
Agribusiness & Commerce : 5,263.41 g
Climate Control : 4,459.96 g

Insight: Furniture and appliance-related categories have the highest average product weights, making them more expensive to store, handle, and deliver. These categories likely contribute disproportionately to logistics and fulfillment costs.

Q8. Which Categories Have the Largest Share of the Product Catalog?

Result:

Bed, Bath & Table : 9.19%
Sports & Leisure : 8.70%
Furniture & Decoration : 8.06%
Beauty & Health : 7.42%
Housewares : 7.09%
Automotive : 5.77%
Computer Accessories : 4.97%
Toys : 4.28%
Watches & Gifts : 4.03%
Telephony : 3.44%

Insight: The marketplace catalog is concentrated in home, lifestyle, and consumer product categories. Bed, Bath & Table alone accounts for over 9% of all listed products, making it the largest category by catalog share.

Q9. How Concentrated Is the Product Catalog?

Result: 62.96%

Insight: The top 10 product categories account for 62.96% of the entire marketplace catalog. This indicates moderate catalog concentration, where a relatively small group of categories contributes the majority of listed products while the remaining categories provide additional diversity.

Q10. How Many Products Have Missing Category Information?

Result:

Products Without Category: 610 (1.85% of all products)

Insight: Only 610 products have missing category information, representing approximately 1.85% of the product catalog. This indicates good overall data quality and ensures that category-level analysis remains reliable for business decision-making.